David Egbakou

Machine Learning Engineer | Data Scientist

Applied Machine Learning • Forecasting • ML Engineering

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Profile

Computer Science engineering student at CY Tech and **Data Scientist (work-study) at DHL Express**. Experience designing, evaluating and automating Machine Learning systems, with a focus on **deep learning, global multi-series forecasting and ensemble learning**. End-to-end owner of a European project covering **39 countries and about 800 facilities**. Interested in ML modeling, model engineering and applied research.

Professional Experience

Data Scientist — Work-Study Program *2025 – 2027*
DHL Express France | Performance & Operations

- **Model design:** built a national forecasting system with **DeepAR**, then redesigned it around **XGBoost**, **STL** seasonal decomposition, dynamic trends and business-driven adjustments.
- **Experimental evaluation:** 5-fold temporal validation; **3.8% WAPE** for the V2 system versus 5.5% for DeepAR V1, on partially overlapping validation periods.
- **ML Engineering:** automated training, retraining and **12-week** recursive inference workflows in **Dataiku DSS**. Model management, Git, K-fold target encoding and publication of forecasts to **Snowflake**.
- **Data Engineering:** Snowflake queries and **PowerShell / RPA** automations feeding a Power BI dashboard that tracks carriers and road delays.

Data Scientist — International Internship — Leipzig, Germany *Summer 2026*
DHL Express Europe

- **End-to-end ownership:** independently designed a forecasting system covering **39 countries and about 800 facilities**, with forecasts at facility and country levels over a **56-day** horizon.
- **Global modeling:** developed and compared multi-series **DeepAR, Chronos-T5 and XGBoost** models, leveraging behavior shared across facilities.
- **Ensemble learning:** combined DeepAR and Chronos-T5 through **gradient-boosted stacking**, complemented by business adjustments.
- **Data robustness:** designed a remapping mechanism for historical parcel volumes to account for changes in facility service areas.
- **Result: 5.6% WAPE** at facility level over a **56-day** horizon. Operational pipeline on **Dataiku DSS and Snowflake**.

Education

Engineering Degree in Computer Science (Data & AI) *2024 – 2027*
CY Tech | Cergy, France

Preparatory Classes MPSI / MP (Mathematics & Physics) *2022 – 2024*
Lycée Technique Errazi | El Jadida, Morocco

Technical Project

Event-Driven ML Architecture — Microservices *October 2026*
Python | Apache Kafka, Docker, Flask, Scikit-learn

- **Distributed architecture:** built containerized microservices with **Docker**, communicating asynchronously through **Apache Kafka** to process business events.
- **Machine Learning integration:** deployed a **RandomForest** inference service, with separate training and inference, Kafka message handling and end-to-end pipeline integration tests.

GitHub — Project source code

Technical Skills

Machine Learning	DeepAR, Chronos-T5, XGBoost, Scikit-learn, Prophet, ensemble learning, time series, STL
ML Methodology	Temporal validation, cross-validation, feature engineering, stacking, model evaluation
Programming	Python, SQL, R, Java, Pandas, NumPy
ML & Data Engineering	Dataiku DSS, Snowflake, Git, ML pipelines, retraining, inference, PostgreSQL
Additional Tools	Power BI, PowerShell, RPA
Languages	French (fluent), English (B2, professional use)